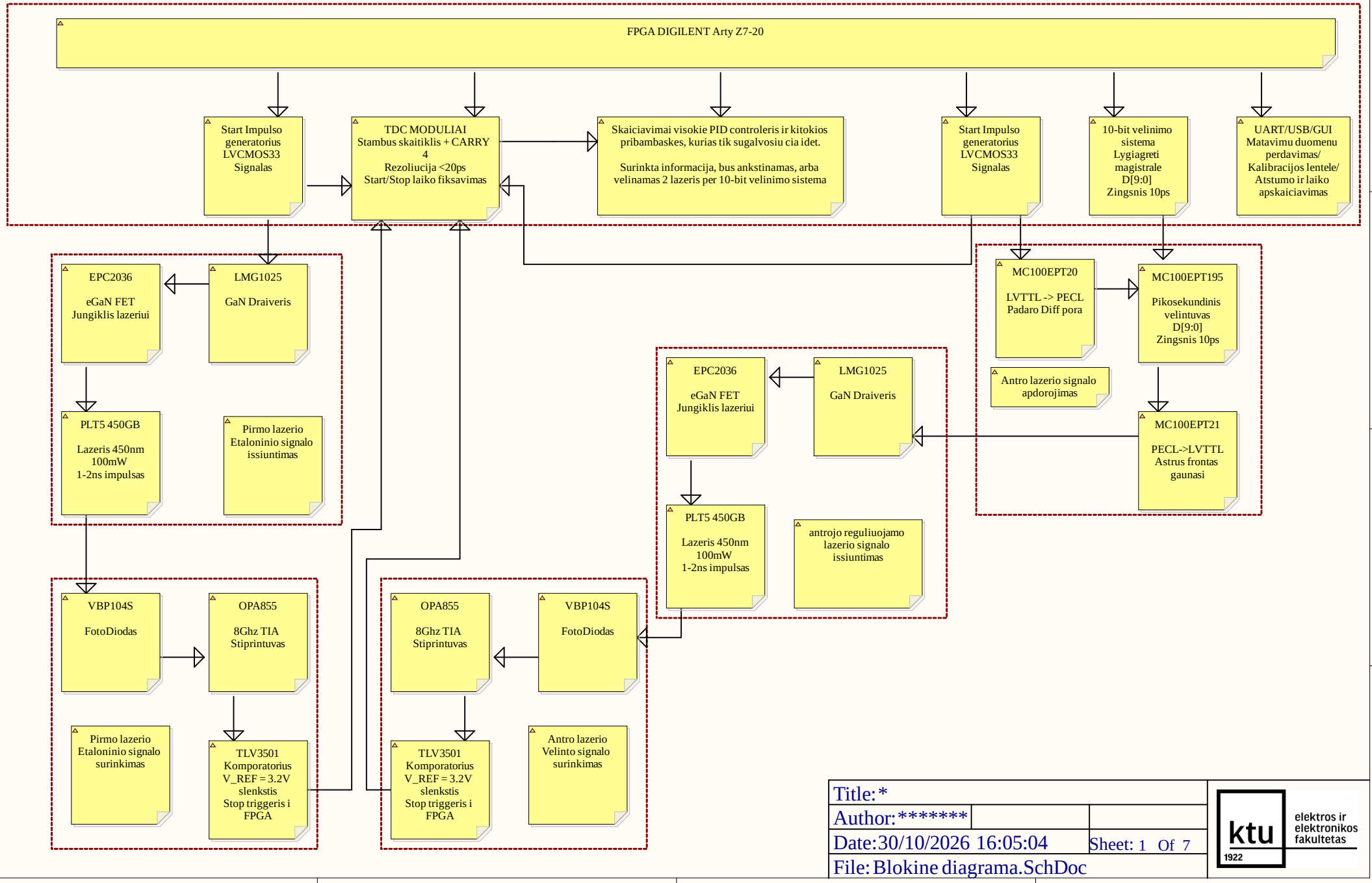

FPGA DIGILENT Arty Z7-20
Start Impulso generatorius LVCMOS33 Signalas
TDC MODULIAI Stambus skaitiklis + CARRY 4 Rezoliucija <20ps Start/Stop laiko fiksavimas
Skaiciavimai visokie PID controleris ir kitokios pribambaskes, kurias tik sugalvosiu cia idet.
Surinkta informacija, bus ankstinamas, arba velinamas 2 lazeris per 10-bit velinimo sistema
Start Impulso generatorius LVCMOS33 Signalas
10-bit velinimo sistema Lygiagreti magistrale D[9:0] Zingsnis 10ps
UART/USB/GUI Matavimu duomenu perdavimas/ Kalibracijos lentele/ Atstumo ir laiko apskaiciavimas
EPC2036 eGaN FET Jungiklis lazeriui
LMG1025 GaN Draiveris
PLT5 450GB Lazeris 450nm 100mW 1-2ns impulsas
Pirmo lazerio Etaloninio signalo issiuntimas
MC100EPT20 LVTTL -> PECL Padaro Diff pora
MC100EPT195 Pikosekundinis velintuvas D[9:0] Zingsnis 10ps
Antro lazerio signalo apdorojimas
MC100EPT21 PECL->LVTTL Astrus frontas gaunasi
EPC2036 eGaN FET Jungiklis lazeriui
LMG1025 GaN Draiveris
PLT5 450GB Lazeris 450nm 100mW 1-2ns impulsas
antrojo reguliuojamo lazerio signalo issiuntimas
VBP104S FotoDiodas
OPA855 8Ghz TIA Stiprintuvas
Pirmo lazerio Etaloninio signalo surinkimas
TLV3501 Komporatorius V_REF = 3.2V slenkstis Stop triggeris i FPGA
OPA855 8Ghz TIA Stiprintuvas
VBP104S FotoDiodas
TLV3501 Komporatorius V_REF = 3.2V slenkstis Stop triggeris i FPGA
Antro lazerio Velinto signalo surinkimas
Title:*
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Matinimas:

Laboratorinis maitinimo saltinis 9V (Galimas maitinimo saltinio triuksmas, ground loop)

2S 18650 8.4V (Visiskai jokio triuksmo ar Ground loopu, bet per laika lazeris silps del krentancios itampos)

LP38798SD-ADJ/ NOPB

+5V RAIL OPA855 & TLV3501

LP38798SD-ADJ/ NOPB

+3V3 RAIL MC100EP195 & SIGNALS

LP38798SD-ADJ/ NOPB

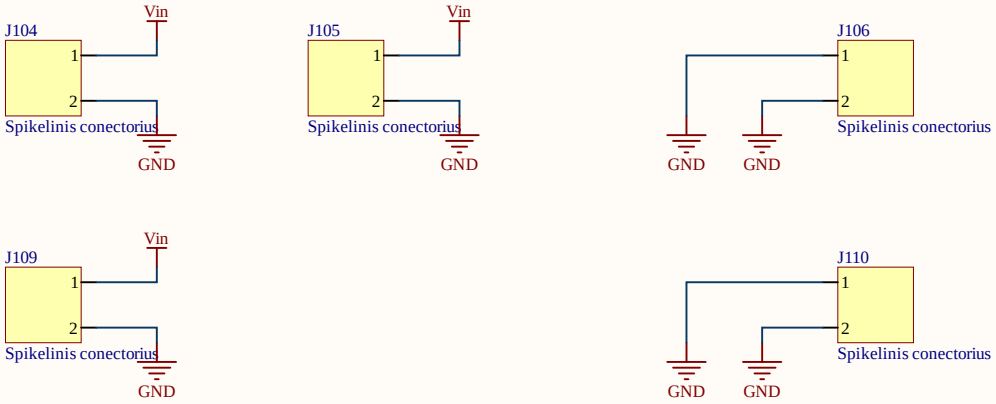
+5V RAIL GaN Driver

Vcc

8.4V 9V RAIL R_charge & C_pulse

J104 Spikelinis jungiklis kad sujungt plokste su Lab Bench PSU 9V Vin

J106 ir J110 Yra skirti papildomiem izeminimam tarp fpga plokstes ir kitu nesamoniu, ka dar sugalvosiu ar galesiu izemint

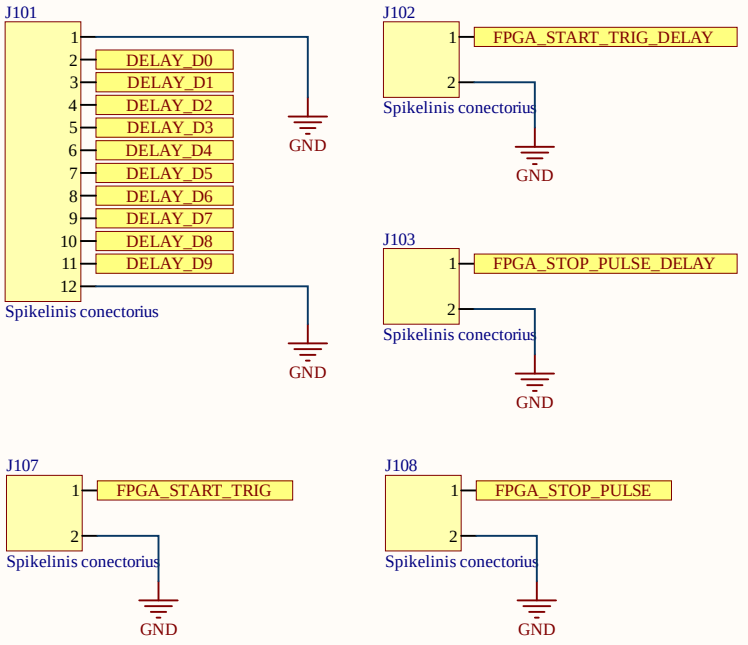


Jungtys is FPGA

J101 Reguliuoja MCP Zingsninio velintojo velinima

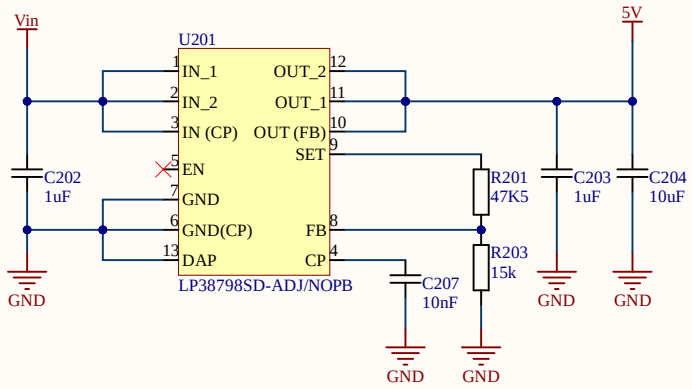
J102 Jungsis su FPGA per PMOD kadangi signalas gan svarbus, tai tures atskyra GND jungti ant pacio pmod

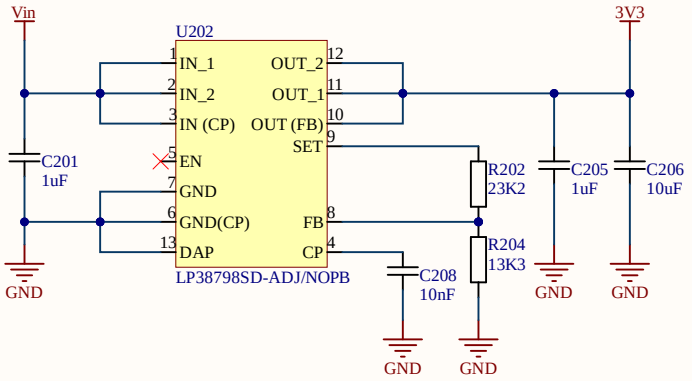
J103 Jungsis su FPGA per PMOD kadangi signalas gan svarbus, tai tures atskyra GND jungti ant pacio pmod

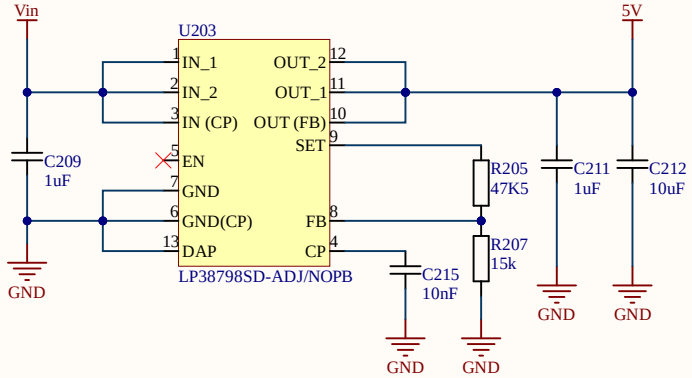


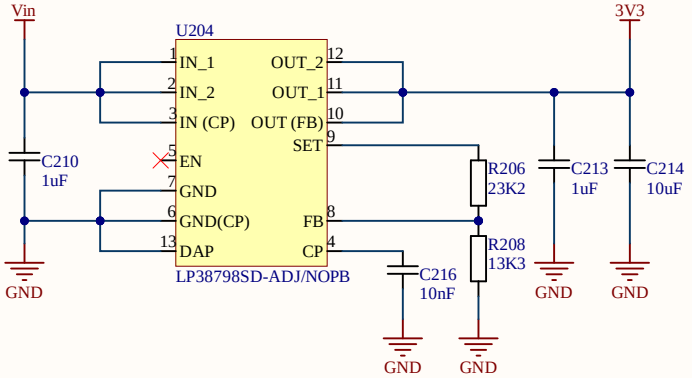
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File: Inputs_Outputs.SchDoc		

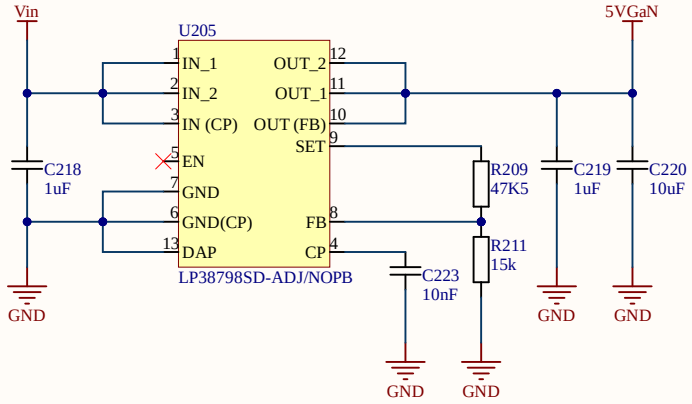
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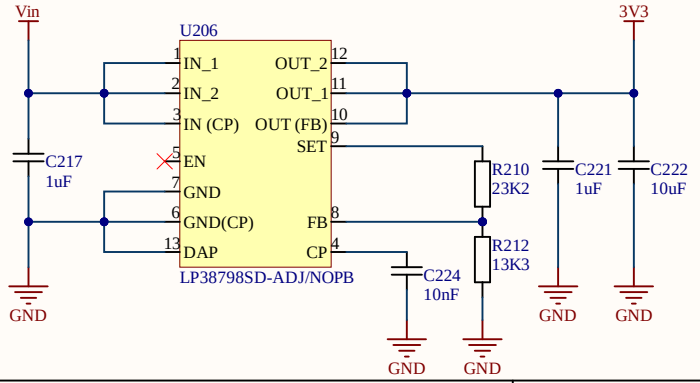












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U401 LVCMOS -> PEVL Vertejas

Ateina signalas is FPGA

U401 pakeicia paprasta itampa i difrencialine pora kad suprastu U403

U403 Velintojas

Ateina signalas is U401

U403 Gali velinti nuo 0ps iki 10230ps (10.23ns) su 10ps zingsniu.

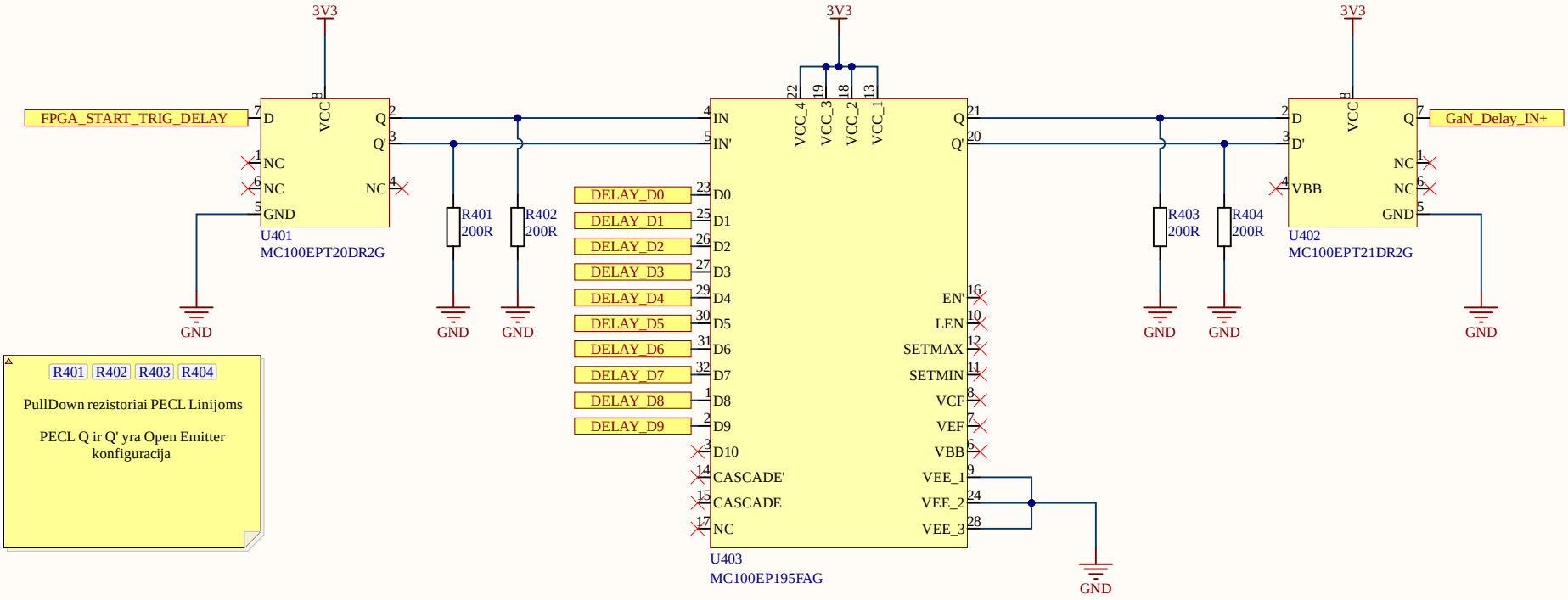
Velinima reguliuoja FPGA per DELAY_D[0:9] 10 bitu magistrale.

Pavelinta signala issiuncia i U402

U402 PEVL -> LVCMOS Vertejas

Ateina diff pora is U403

U402 pakeicia difrencialine pora i paprasta signala kad suprastu U301

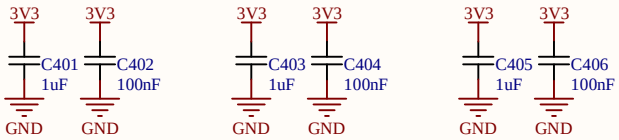


R401 R402 R403 R404

PullDown rezistoriai PECL Linijoms

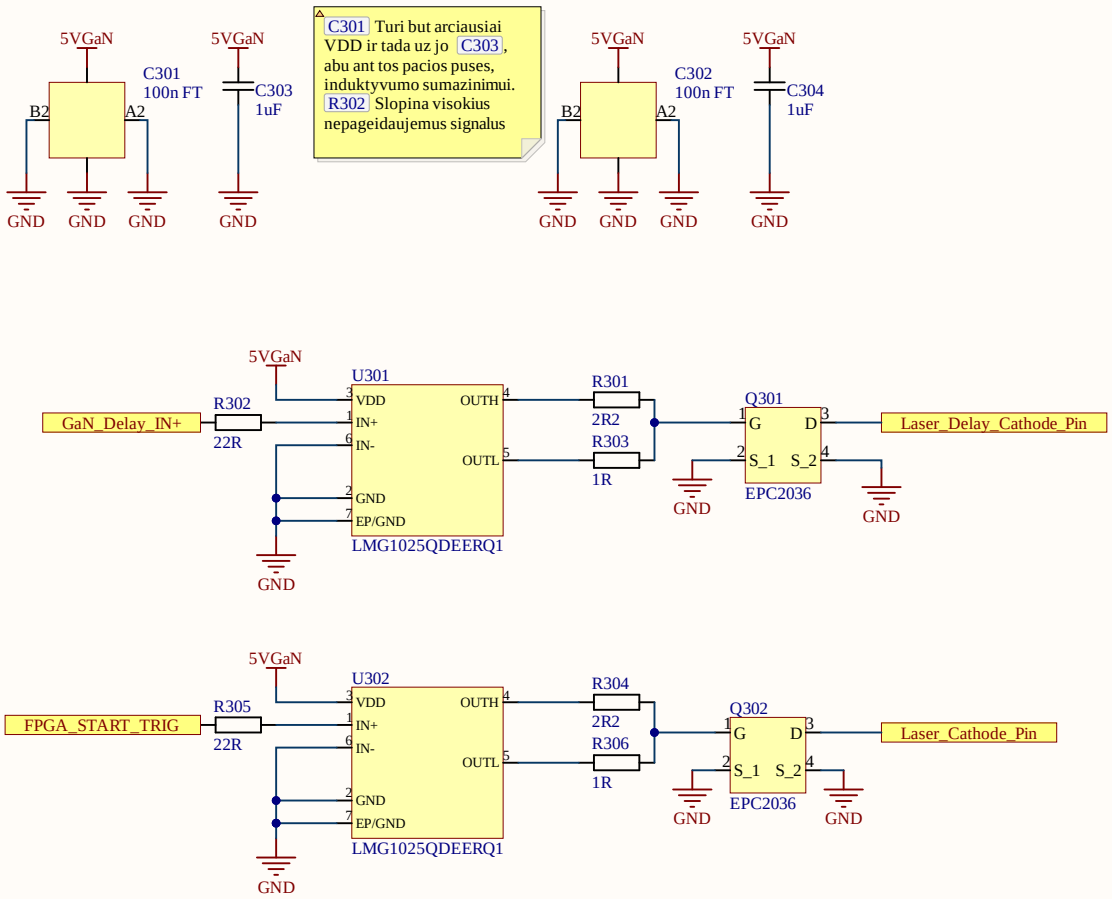
PECL Q ir Q' yra Open Emitter konfiguracija

C401 - C406
Bus prie U401 - U403 Iejimu



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File: Time_Delay.SchDoc		

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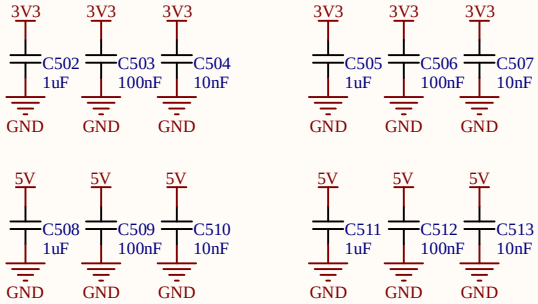


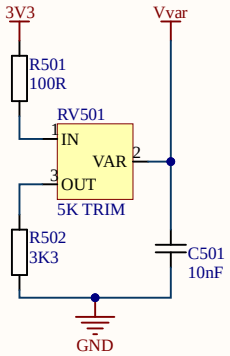
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Author:*******		
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C502 - C504 ir C508 - C513

Reikalingi filtravimui, bei stabilios itampos islaikymui.
Kiekviena eile kondensatoriu lituojasi kuo arciau mikruskes
U501 - U504





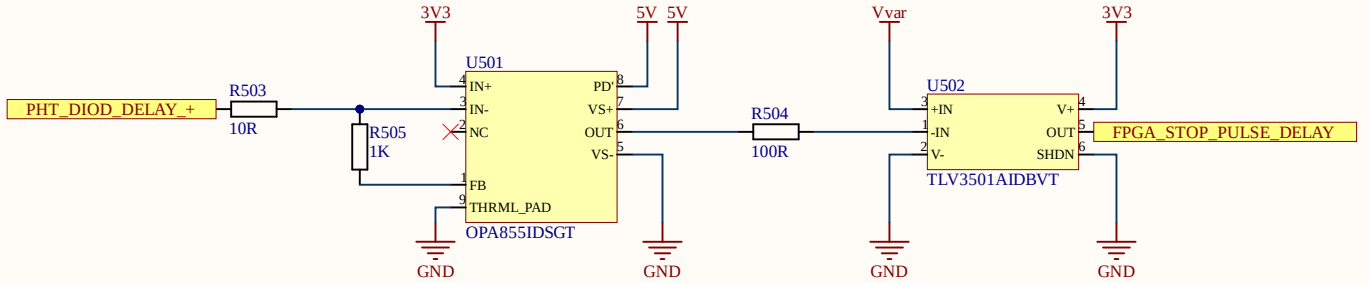
R501 R502 Apriboja bendra srove teknacia itampos daliklyje
RV501 Leidzia reguliuoti itampa
Nuo 1.30V Iki 3.25V

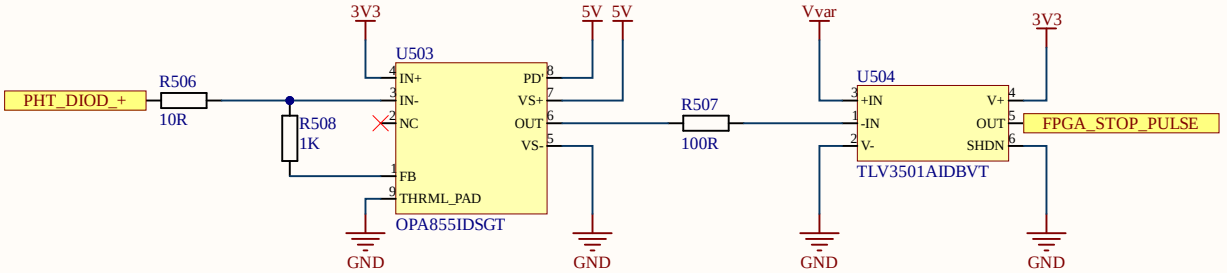
Normali busena:
Fotodiodo srove Ipd = 0mA (tamsu) Vout = 3.3V

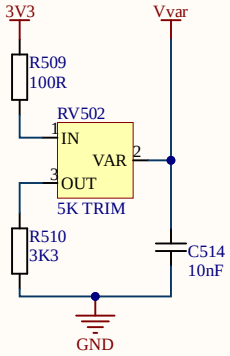
Apsviesta busena:
Fotodiodo srove Ipd = 1mA (blyksnis) Vout = 3.3V - 1mA * 1k = 2.3V
(Dar neskaiciaviau kokias sroves gali paduot fotodiodas, nuo to keisis FB rezistorius)

U501 Sustiprina Fotodiodo generuojama signala, kuris pasiekes itampa < Vvar pakeicia U502 komporatoriaus verte i logini 1 (3V3) kuri supranta FPGA.

RV501 Reguliuoja Vvar Itampa
absoliucioj tamsoj gal tiks ir 3.25V slenkstis
Apsviestam kambari gali but reikes ir 1.30V slenkscio







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R601 R603 ir C601 C603 Talpinė lazerių iškrova

- R601 / R603 (1k) krauna kondensatorius C601 ir C603 (2nF) iki 9V.

- Atsidarius GaN tranzistoriui, kondensatorius akimirksniu išsikrauna per lazerį (D601 / D603) į žemę ir duoda aštrų sub-ns šviesos impulsą.

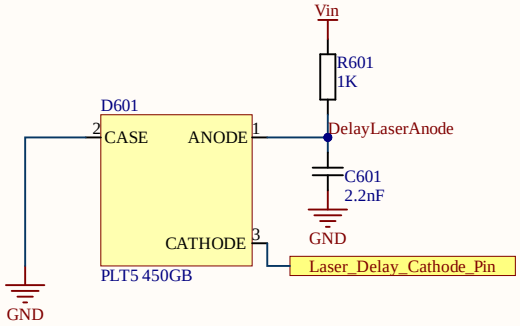
- Kondensatoriui išsikrovus lazeris užgęsta pats, o 1k varža apsaugo jį nuo sudegimo.

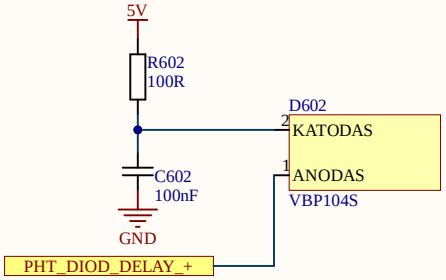
R602 R604 ir C602 C604 Fotodiodų priešįtampis

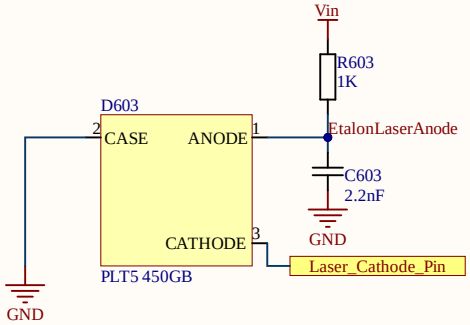
- R602 / C602 ir R604 / C604 RC filtrai išvalo 5V maitinimo triukšmus.

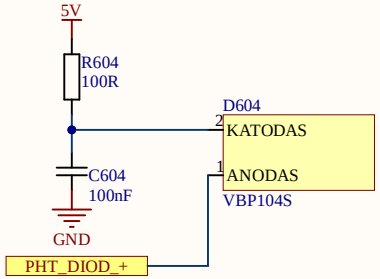
- Katodai pajungti prie 5V, o Anodai per OPA855 (U501 ir U503) sėdi ties 3.3V (1.7V atbulinis priešįtampis maksimaliai sumažina diodo talpą ir padidina greitį).

- Šviesos blyksnio srovė siunčiama tiesiai į OPA855 stiprintuvą.



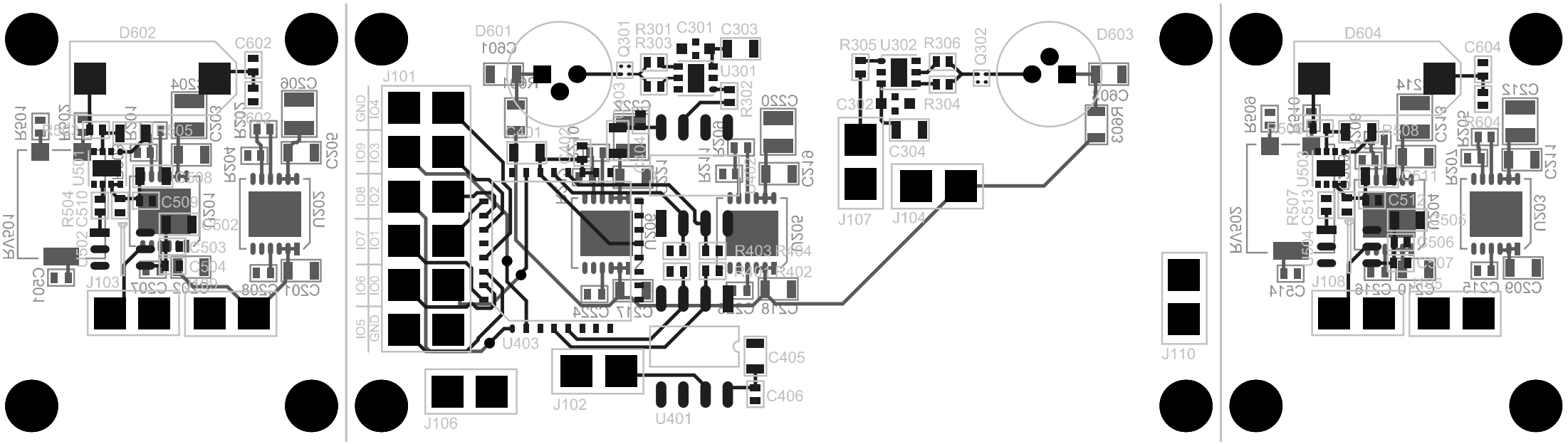






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D602
C602
J103
J101
D601
R301
C301
C303
U301
R303
Q301
R302
C220
C218
R403
R404
R402
U403
C405
C406
J102
U401
J106
R305
U302
R306
Q302
D603
C302
R304
C304
J107
J104
J110
D604
C604
J108
C514
C509
C503
C504
C502
C507
C513
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